Justin Gouge

Computer Scientist

+12043308486 ◇ gougej@yorku.ca ◇ Toronto, Ontario, Canada ◇ Open to Relocate ◇ LinkedIn ◇ Portfolio

SUMMARY

IEEE & Springer published author and machine learning researcher with a strong focus on federated learning and distributed learning over the cloud. Experienced in developing novel algorithms to enhance system resilience, analyze attack scenarios, and optimize model performance. Seeking to leverage expertise in AI/ML and distributed learning to advance secure and scalable machine learning solutions.

EXPERIENCE

Prompt Engineer — Nov '24 — Present
Outlier — Toronto, Canada

- Design, test, and optimize AI prompts to enhance model performance, ensuring high-quality and contextually relevant responses for large language models.
- Collaborate with AI researchers and engineers to refine large language models, leveraging prompt engineering techniques to improve accuracy, efficiency, and user experience.
- Analyze model outputs and user feedback to develop prompt frameworks, automate prompt evaluation, and contribute to the continuous improvement of AI-driven solutions.

Research Assistant — Sep '21 — Apr '24
York University — Toronto, Canada

- Pioneered novel algorithms for detecting malicious entities within federated learning systems, decreasing false positives by 40% and improving the overall system's security and reliability.
- Analyzed 300+ research publications on machine learning methodologies, identifying key trends and gaps in the field of federated learning, leading to a focused research direction.
- Evaluated 5 distinct algorithms, including differential privacy and secure aggregation, to build a simulated system, identifying three key performance bottlenecks that limit scalability in real-world deployments.
- Applied novel algorithms to 3 different attack scenarios to measure performance against 3 different benchmark scenarios.

Teaching Assistant — Sep '21 — Apr '24
York University — Toronto, Canada

- Cultivated an inclusive educational setting for computer science students, orchestrating and directing 2+ weekly lab sessions; deepening the comprehension of complex algorithms & data structures for students.
- Mentored 25+ students weekly in algorithms and software design, resulting in a 90% pass rate on related assessments and improved understanding of complex computer science principles using detailed tailored feedback.
- Led interactive group discussions on advanced software design principles, fostering a collaborative learning environment that resulted in a 90% student participation rate and deeper understanding of course material.

Security Engineer — Jan '21 — Aug '21
Communications Security Establishment — Ottawa, Canada

- Engineered enhancements to the organization's applicant system with agile methodologies, resolving 100+ tickets and decreasing applicant time by 15% while improving application security posture across the organization.
- Orchestrated cross-functional collaboration with developers, system designers, and analysts, integrating security protocols into feature planning; decreased application vulnerabilities and fortified system resilience.
- Implemented a full-stack applicant tracking system using Python, Java and SQL, automating new-hire security clearance screenings that reduced processing time by 50%.

Software Developer — Sep '18 — Dec '18
Communications Security Establishment — Ottawa, Canada

- Examined and verified the feasibility of hosting an open-source social network, presenting findings to fix the 3 biggest causes of crashes, thereby preventing potential service disruptions and improving overall system reliability.
- Constructed a Django web platform from scratch specifically for cybersecurity vulnerability analysis, completing the project within 6 weeks while meeting all design requirements.
- Designed a cross-platform polling application compatible with Android and Windows, facilitating real-time data collection, and enhancing decision-making speed by 30% for project stakeholders, improving communication.

Software Developer — Jan '18 — Apr '18
Varian Medical Systems — Winnipeg, Canada

- Completed targeted front-end bug fixes reported by user feedback channels which bolstered application functionality; also, enhanced back-end data handling by integrating 100+ lines of code.
- Drove the planning and implementation of 5 user-requested features within the application, while personally resolving 10 15 bug tickets weekly, maintaining responsiveness to pressing user needs.
- Rectified 95% of reported linguistic inconsistencies within the application's multilingual interface, guaranteeing uniform and precise messaging across 2 distinct language settings throughout the term.

EDUCATION

Master of Computer Science, York University — Sep '21 — Feb '25, Toronto, Canada

Bachelor of Computer Science, University of Manitoba — Sep '15 — Dec '20, Winnipeg, Canada

PROJECTS & PUBLICATIONS

Securing Multi-Layer Federated Learning: Detecting and Mitigating Adversarial Attacks Link — Sep '24

- Introduced new methods for malicious client detection in multi-layer federated learning systems.
- Implemented 2 novel approaches for removing clients to prevent negative effects on the model's learning process.
- Benchmarked new advancements against previous work and demonstrated noticeable improvements.

Mining for Fake News Link — Mar '22

- Evaluated the performance of different machine learning models for fake news detection.
- Employed Sequential mining and bag-of-words (BoW) to classify articles as real or fake.
- Experimented with 3 datasets to capture widespread scenarios and conducted experiments to test the accuracy of fake news detection algorithms.

Enhanced Sliding Window-Based Periodic Pattern Mining from Dynamic Streams Link — Apr '20

- Proposed improvements to existing insertion and deletion algorithms for suffix trees.
- Developed a suite of optimized suffix tree operations, achieving an improvement in query processing speed compared to existing methods when handling complex time-series datasets.

3D Music Visualizer Link — Dec '18

- Created software that visualizes music with 5+ scenes and hundreds of 3D shapes.

Federated Learning User Verification - Code Reproduction Link

- Spearheaded a comprehensive reproducibility study of a published machine learning paper, rigorously verifying the original authors' claims and computational experiments through an independent implementation.
- Authored a detailed report analyzing the reproducibility of the original paper, including any discrepancies and providing insights into the robustness of the published results.
- Gained hands-on experience with state-of-the-art machine learning models and techniques while contributing to the broader scientific community's efforts to ensure the reliability and validity of research in the field.

Firelight Duel Academy Link

- Developed a full-stack Yu-Gi-Oh! TCG e-commerce platform with Node.js/Express backend, MySQL database (14+ tables), JWT authentication, and comprehensive REST API supporting user management, shopping cart, orders, and payment processing with Square integration
- Built responsive frontend with modern JavaScript (ES6+) featuring single-page application architecture, multiple API integrations (YGOProDeck, TCGPlayer, Crystal Commerce, PriceCharting), advanced search functionality, and mobile-optimized UI with 130+ curated card images.
- Implemented production-ready DevOps pipeline including GitHub Actions CI/CD with automated testing, code quality checks, security scanning, comprehensive documentation, and deployment to GitHub Pages with support for multiple hosting environments.

SKILLS

Relevant Skills OpenAI, Technology, Product, Visionary, Agents
Technical Skills Python, Java, C, C++, C#, Ruby, Assembly, Html, CSS, SQL, JavaScript, TypeScript
Data Skills Data Analysis, Data-Driven Decision-Making, Data Visualization, Business Intelligence
Development Tools PyTorch, NumPy, Visual Studio, Visual Studio Code, Abode Suite, Android Studio, IntelliJ, Postman, Microsoft SQL server, Git, GitHub, Gerrit
Web Development Django, Angular JS
Methodologies Machine Learning, Deep Learning, Federated Learning, Modeling, Artificial Intelligence